A complete proof draft of the Hautus criterion for normal generators with horizontal-strip spectrum

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Status and scope: complete proof draft. The argument addresses the full horizontal-strip implication stated in Theorem 1, including infinite-dimensional Hilbert output and continuous normal spectrum. All required lemmas are proved below; no unresolved mathematical step is knowingly deferred. This status describes the scope and completeness of the proposed argument, not external certification: the draft has received internal adversarial checks but has not been independently peer reviewed or formally verified. General sector spectra are outside the stated problem and are not claimed here.

Abstract

We give an affirmative proof of the sufficiency of the quantitative Hautus condition for infinite-time exact observability of an exponentially stable normal semigroup whose generator has bounded imaginary spectrum. The output space may be infinite dimensional and the generator may have continuous spectrum. The main step shows that every unit sequence with vanishing observation energy must lose all spectral mass in bounded subsets of the strip. This follows from complex-time analyticity, scalar normal-family compactness, and exponential concentration at an exposed point of a limiting spectral measure. A fixed exponential-polynomial filter then gives a contradiction, because the spectral angle tends to zero at the left end of a horizontal strip. The filter is adapted from the self-adjoint argument of Daun and Jacob [1, Section 4]; the proposed additional step is the local spectral escape argument.

1 Precise statement and conventions

All Hilbert spaces are complex. We write $\mathbb{C}_+ = \{w \in \mathbb{C} : \operatorname{Re} w > 0\}$ and $\mathbb{C}_- = \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta < 0\}$. The graph norm on $D(A)$ is $\|x\| + \|Ax\|$.

Theorem 1 (Horizontal-strip sufficiency)**.** *Let A be a normal generator of an exponentially stable strongly continuous semigroup $T(t)$ on a Hilbert space X. Suppose*

$$\sup_{\lambda\in\sigma(A)} |\operatorname{Im}\lambda| < \infty.$$

Let Y be a Hilbert space, and let $C : D(A) \to Y$ be linear, bounded for the graph norm, and L^2-admissible. Suppose that some $m > 0$ satisfies

$$\|(\zeta - A)x\|^2 + |\operatorname{Re}\zeta|\,\|Cx\|^2 \geq m\,|\operatorname{Re}\zeta|^2\,\|x\|^2, \qquad \zeta \in \mathbb{C}_-, \quad x \in D(A). \tag{H}$$

Then there exists $c > 0$ such that

$$\mathcal{E}(x) := \int_0^\infty \|CT(t)x\|^2\,\mathrm{d}t \geq c\,\|x\|^2, \qquad x \in D(A). \tag{O}$$

The same lower bound holds on X for the admissible observation map.

Exponential stability gives a spectral gap. Together with the strip hypothesis, we may fix $a > 0$ and $B \geq 0$ such that

$$\sigma(A) \subset S := \{\lambda = u + iv : \ u \leq -a, \ |v| \leq B\}. \tag{1.1}$$

By the spectral theorem for normal operators, $\|T(t)\| \leq e^{-at}$. If admissibility is initially assumed on an interval $[0, t_0]$, with upper bound $M_0 \|x\|^2$, applying it to $T(kt_0)x$ and summing over $k \geq 0$ gives

$$\mathcal{E}(x) \leq M \|x\|^2, \qquad M = \frac{M_0}{1 - e^{-2at_0}}. \tag{1.2}$$

Thus finite-time and infinite-time admissibility are interchangeable here.

Let E_A be the projection-valued spectral measure of A. For $x \in X$, define the finite positive measure

$$\nu_x(\Omega) = \|E_A(\Omega)x\|^2 \tag{1.3}$$

on S. Its mass is $\|x\|^2$. We use arbitrary spectral measures; no basis of eigenvectors is assumed.

Daun and Jacob [1, Section 6] formulate the horizontal-strip case as an open direction after proving self-adjoint sufficiency and exhibiting a counterexample for general normal generators. Their Section 4 supplies the exponential-polynomial filter used below. The new assertion proposed in this draft is the escape lemma and its combination with that filter for a strip.

2 Vanishing observation at fixed complex times

For $w \in \mathbb{C}_+$, normal functional calculus defines $T(w) = e^{wA}$. For every integer $r \geq 0$ and every compact $K \subset \mathbb{C}_+$,

$$\sup_{w \in K} \sup_{\lambda \in S} |\lambda|^r \left| e^{w\lambda} \right| < \infty. \tag{2.1}$$

Indeed, write $w = \delta + i\eta$ and $\lambda = -h + ib$. Then

$$\left| e^{w\lambda} \right| = e^{-\delta h - \eta b} \leq e^{-\delta h + B|\eta|}, \qquad h \geq a, \quad |b| \leq B.$$

On K, δ is bounded below by a positive number and $|\eta|$ is bounded. Exponential decay in h controls every fixed power of $|\lambda|$. In particular, $T(w)X \subset D(A)$, and $w \mapsto T(w)$ is holomorphic as an operator from X to $D(A)$ with its graph norm. This follows by differentiating spectral multipliers; (2.1) also controls the graph norms of the derivatives.

Lemma 2 (Complex-time vanishing)**.** *Let $x_n \in D(A)$ satisfy $\|x_n\| \leq 1$ and $\mathcal{E}(x_n) \to 0$. Then*

$$\|CT(w)x_n\| \longrightarrow 0 \qquad \textit{for each fixed } w \in \mathbb{C}_+. \tag{2.2}$$

Proof. Set $F_n(w) = CT(w)x_n$. The graph-norm boundedness of C and (2.1) make these Y-valued holomorphic functions locally uniformly bounded on $\mathbb{C}_+$.

If the conclusion fails at w_0, pass to a subsequence such that $\|F_n(w_0)\| \geq \varepsilon > 0$. Choose norm-one complex-linear functionals $\ell_n \in Y^*$ with $|\ell_n(F_n(w_0))| = \|F_n(w_0)\|$. The scalar holomorphic functions $g_n = \ell_n \circ F_n$ are locally uniformly bounded. Scalar Montel's theorem therefore supplies a further subsequence converging locally uniformly to a holomorphic function g on $\mathbb{C}_+$.

On any fixed real interval $[r, s] \subset (0, \infty)$,

$$\int_r^s |g_n(t)|^2 \, \mathrm{d}t \leq \mathcal{E}(x_n) \longrightarrow 0.$$

Local uniform convergence implies $\int_r^s |g(t)|^2 \, \mathrm{d}t = 0$. Continuity, then the identity theorem, give $g \equiv 0$. This contradicts $|g_n(w_0)| \geq \varepsilon$. Notice that compactness was used only for scalar holomorphic functions, not for an infinite-dimensional vector-valued family. □

3 The local spectral escape lemma

Lemma 3 (Escape from every bounded part of the strip)**.** *Under* (1.1) *and* (H)*, let* $x_n \in D(A)$*,* $\|x_n\| \le 1$*, and* $\mathcal{E}(x_n) \to 0$*. For every* $\tau > 0$*,*

$$\|T(\tau)x_n\| \longrightarrow 0. \tag{3.1}$$

Consequently, for every finite $R \ge a$*,*

$$\nu_{x_n}\{\lambda \in S : -\operatorname{Re}\lambda \le R\} \longrightarrow 0. \tag{3.2}$$

Proof. We prove (3.1) by contradiction in four steps.

Step 1: a nonzero tight measure limit. Suppose that for some $\tau > 0$, after passing to a subsequence, $\|T(\tau)x_n\| \ge \eta > 0$. Write $\nu_n = \nu_{x_n}$ and define

$$\mathrm{d}\mu_n(\lambda) = e^{2\tau \operatorname{Re}\lambda}\,\mathrm{d}\nu_n(\lambda). \tag{3.3}$$

Then $\eta^2 \le \mu_n(S) \le 1$, and, for $R \ge a$,

$$\mu_n\{-\operatorname{Re}\lambda > R\} \le e^{-2\tau R}.$$

The bounded imaginary coordinate makes this a tight family of finite measures on the closed subset $S \subset \mathbb{R}^2$. First pass to a subsequence whose total masses converge. Apply Prokhorov's theorem to the probability measures $\mu_n/\mu_n(S)$, which remain tight since $\mu_n(S) \ge \eta^2$. Thus a subsequence of μ_n converges weakly to a finite measure μ with $\mu(S) \ge \eta^2 > 0$. Moreover,

$$\int_S |\lambda|^2\,\mathrm{d}\mu_n(\lambda) \le \sup_{h \ge a}(h^2 + B^2)e^{-2\tau h} < \infty.$$

Applying weak convergence to the bounded continuous functions $\min\{|\lambda|^2, L\}$, then letting $L \to \infty$, shows

$$\int_S |\lambda|^2\,\mathrm{d}\mu(\lambda) < \infty. \tag{3.4}$$

Step 2: pass the Hautus inequality to the measure. Fix $w \in \mathbb{C}_+$ and $\zeta \in \mathbb{C}_-$. Apply (H) to $T(\tau + w)x_n \in D(A)$. By Lemma 2, its observation term tends to zero. The spectral theorem and (3.3) give, after taking the weak limit,

$$\int_S |\zeta - \lambda|^2 e^{2\operatorname{Re}(w\lambda)}\,\mathrm{d}\mu(\lambda) \ge m|\operatorname{Re}\zeta|^2 \int_S e^{2\operatorname{Re}(w\lambda)}\,\mathrm{d}\mu(\lambda). \tag{3.5}$$

For each fixed $w \in \mathbb{C}_+$, both integrands are bounded and continuous on S: the decay along $\operatorname{Re}\lambda \to -\infty$ dominates the quadratic factor. This justifies the weak-limit passage without any unbounded test function. The resulting inequality holds for every fixed w, ζ.

Step 3: find a unique exposed point of the support. Let $F = \operatorname{supp}\mu \subset S$, and define

$$h(q) = \sup_{u+iv \in F}(u + qv), \qquad q \in \mathbb{R}. \tag{3.6}$$

The nonempty set F, the upper bound $u \le -a$, and $|v| \le B$ show that h is finite, convex, and B-Lipschitz. Choose a point q_0 where h is differentiable, which exists by elementary one-dimensional convex analysis. The supremum in (3.6) is attained: the quantity $u + q_0 v$ tends uniformly to $-\infty$ as $u \to -\infty$, so a maximizing sequence lies in a compact subset of the closed set F.

If $u+iv$ is any maximizer at q_0, then $h(q) \geq h(q_0) + (q-q_0)v$ for all q. Thus v is a subgradient of h at q_0, and differentiability forces $v = h'(q_0)$. The equation $u + q_0 v = h(q_0)$ then also fixes u. Consequently there is exactly one maximizer, denoted by $\lambda_0 \in F$. In particular, $\operatorname{Re}\lambda_0 \leq -a < 0$.

Step 4: exponential concentration contradicts (3.5). Put $\phi(u+iv) = u + q_0 v$ and $H_0 = \phi(\lambda_0)$. In (3.5), choose $w = t(1 - iq_0)$ and $\zeta = \lambda_0$, with $t > 0$. Both choices are in their required half-planes. Division by a positive denominator yields

$$\frac{\int_S |\lambda - \lambda_0|^2 e^{2t\phi(\lambda)}\, \mathrm{d}\mu(\lambda)}{\int_S e^{2t\phi(\lambda)}\, \mathrm{d}\mu(\lambda)} \geq m |\operatorname{Re}\lambda_0|^2 \geq m a^2. \tag{3.7}$$

We show that the left side tends to zero. Fix $\varepsilon > 0$. Uniqueness of the maximizer and the decay of ϕ at the left end of the strip imply a strict gap: if the complement of the ball is nonempty, there is $\delta > 0$ such that

$$\sup_{\lambda \in F,\ |\lambda - \lambda_0| \geq \varepsilon} \phi(\lambda) \leq H_0 - \delta.$$

This follows by taking a convergent subsequence of any sequence that would violate the gap. The set $U = \{\lambda \in S : \phi(\lambda) > H_0 - \delta/2\}$ has positive μ-measure, since $\lambda_0 \in \operatorname{supp}\mu$. By (3.4), the contribution from outside the ball to the quotient in (3.7) is bounded by

$$\frac{\int_S |\lambda - \lambda_0|^2\, \mathrm{d}\mu(\lambda)}{\mu(U)} e^{-t\delta} \longrightarrow 0. \tag{3.8}$$

The contribution from inside the ball is at most ε^2. If the complement is empty, that estimate alone suffices. Letting first $t \to \infty$ and then $\varepsilon \downarrow 0$ contradicts (3.7). This proves (3.1).

Finally, for every fixed R and $\tau > 0$, the spectral theorem gives

$$\nu_n\{-\operatorname{Re}\lambda \leq R\} \leq e^{2\tau R} \int_S e^{2\tau \operatorname{Re}\lambda}\, \mathrm{d}\nu_n(\lambda) = e^{2\tau R} \|T(\tau)x_n\|^2 \longrightarrow 0.$$

This is (3.2). □

Remark (Order of limits). The limit $n \to \infty$ is taken at fixed w and ζ, before the concentrating parameter $t \to \infty$ is introduced. No bound on $CT(w)$ uniform over an unbounded set of complex times is needed.

4 An exponential-polynomial filter

The construction in this section starts from the filter in [1, Propositions 4.1–4.3]; we include the details needed for complex spectral rays. Write $L = \log 2$, and, for $N \geq 1$, set

$$f_N(w) = \left[e^{-Lw}(1 - e^{-Lw})\right]^N = \sum_{j=0}^{N} (-1)^j \binom{N}{j} e^{-\beta_j w}, \qquad \beta_j = (N+j)L > 0. \tag{4.1}$$

For $q \in \mathbb{R}$, define

$$a_N(q) = \int_0^\infty |f_N(s(1 - iq))|^2 \frac{\mathrm{d}s}{s}, \tag{4.2}$$

$$b_N(q) = \int_0^\infty |1 - s(1 - iq)|^2 |f_N(s(1 - iq))|^2 \frac{\mathrm{d}s}{s}. \tag{4.3}$$

Lemma 4 (Continuity and concentration of the filter)**.** *For every finite $Q \geq 0$, a_N and b_N are finite, continuous, and bounded on $[-Q, Q]$. Moreover,*

$$a_N(0) > 0, \qquad \frac{b_N(0)}{a_N(0)} \longrightarrow 0 \quad \text{as } N \to \infty. \tag{4.4}$$

Proof. When $w = s(1 - iq)$, $s > 0$, the identity $1 - e^{-Lw} = w \int_0^L e^{-rw}\,\mathrm{d}r$ and the triangle inequality give $\left|1 - e^{-Lw}\right| \le \min\{L|w|, 2\}$. Thus, for $|q| \le Q$,

$$|f_N(s(1 - iq))| \le e^{-NLs} \min\{Ls\sqrt{1 + Q^2}, 2\}^N, \tag{4.5}$$
$$|1 - s(1 - iq)| \le 1 + s\sqrt{1 + Q^2}.$$

The resulting dominating functions in (4.2)–(4.3) behave as $O(s^{2N-1})$ near zero and as a polynomial times e^{-2NLs} at infinity. Dominated convergence proves the first assertion.

For the ratio, write $g(s) = e^{-Ls}(1 - e^{-Ls})$. This is positive on $(0, \infty)$, tends to zero at both ends, and has the unique maximum $g(1) = 1/4$. Fix $0 < \varepsilon < 1$, and let

$$r = \sup_{|s-1| \ge \varepsilon} g(s) < 1/4.$$

Choose $p \in (r, 1/4)$ and an interval J about 1, contained in $\{|s - 1| < \varepsilon\}$, on which $g \ge p$. Then

$$a_N(0) \ge p^{2N} \int_J \frac{\mathrm{d}s}{s}.$$

The part of $b_N(0)$ with $|s - 1| < \varepsilon$ is at most $\varepsilon^2 a_N(0)$. On its complement, use $g^{2N} \le r^{2N-2} g^2$ to obtain

$$\frac{b_N(0)}{a_N(0)} \le \varepsilon^2 + \frac{r^{2N-2} b_1(0)}{p^{2N} \int_J \mathrm{d}s/s}.$$

Since $r/p < 1$, the last term tends to zero. Now let $\varepsilon \downarrow 0$, proving (4.4). □

Proposition 5 (Integrated Hautus estimate on the strip)**.** *For $\lambda = -h + ib \in S$, put $q(\lambda) = b/h$. For every $N \ge 1$ and $x \in D(A)$,*

$$\int_S \left[m a_N(q(\lambda)) - b_N(q(\lambda))\right] \mathrm{d}\nu_x(\lambda) \le \kappa_N \mathcal{E}(x), \tag{4.6}$$

where the finite constant is

$$\kappa_N = \left(\sum_{j=0}^{N} \frac{\binom{N}{j}}{\sqrt{(N + j)\log 2}} \right)^2. \tag{4.7}$$

Proof. Fix $t > 0$. By (4.1),

$$v_t = f_N(-tA)x = \sum_{j=0}^{N} (-1)^j \binom{N}{j} T(\beta_j t)x \in D(A).$$

Set $\zeta = -1/t$ and $v = v_t$ in (H), multiply by t, and integrate in t. This yields

$$m I_1(x) \le I_2(x) + \int_0^\infty \|C f_N(-tA)x\|^2 \,\mathrm{d}t, \tag{4.8}$$

where

$$I_1(x) = \int_0^\infty \|f_N(-tA)x\|^2 \,\frac{\mathrm{d}t}{t},$$
$$I_2(x) = \int_0^\infty \|(I + tA) f_N(-tA)x\|^2 \,\frac{\mathrm{d}t}{t}.$$

The spectral theorem and Tonelli's theorem apply to these nonnegative integrands. At $\lambda = -h + ib$, substitute $s = th$, so that $-t\lambda = s(1 - iq(\lambda))$. It follows that

$$I_1(x) = \int_S a_N(q(\lambda))\,\mathrm{d}\nu_x(\lambda), \qquad I_2(x) = \int_S b_N(q(\lambda))\,\mathrm{d}\nu_x(\lambda). \tag{4.9}$$

Since $|q(\lambda)| \le B/a$, Lemma 4 proves that both integrals are finite.

Finally, the triangle inequality in $L^2((0,\infty); Y)$, followed by time scaling, gives

$$\begin{aligned}\left(\int_0^\infty \|Cf_N(-tA)x\|^2\,\mathrm{d}t\right)^{1/2} &\le \sum_{j=0}^{N}\binom{N}{j}\left(\int_0^\infty \|CT(\beta_j t)x\|^2\,\mathrm{d}t\right)^{1/2} \\ &= \sqrt{\kappa_N}\,\mathcal{E}(x)^{1/2}.\end{aligned} \tag{4.10}$$

Combine (4.8)–(4.10) to obtain (4.6). At no point is C commuted with a spectral projection or with the imaginary part of A. □

5 Completion of the proof

Proof of Theorem 1. Suppose that (O) fails. There are $x_n \in D(A)$ such that

$$\|x_n\| = 1, \qquad \mathcal{E}(x_n) \longrightarrow 0. \tag{5.1}$$

Let $\nu_n = \nu_{x_n}$; these are probability measures on S. Lemma 3 says that for every finite $R \ge a$, $\nu_n\{h \le R\} \to 0$, where $h = -\operatorname{Re}\lambda$. Since $|\operatorname{Im}\lambda| \le B$, the ratio $q(\lambda) = \operatorname{Im}\lambda/(-\operatorname{Re}\lambda)$ tends to zero in ν_n-probability. Explicitly, for $B > 0$ and $\varepsilon > 0$,

$$\nu_n\{|q| \ge \varepsilon\} \le \nu_n\{h \le B/\varepsilon\} \longrightarrow 0;$$

when $B/\varepsilon < a$, the set on the left is empty. For $B = 0$, the assertion is immediate.

By Lemma 4, fix one N so large that

$$d_0 := ma_N(0) - b_N(0) > 0. \tag{5.2}$$

The function $d(q) = ma_N(q) - b_N(q)$ is bounded and continuous on $[-B/a, B/a]$. Therefore

$$\int_S d(q(\lambda))\,\mathrm{d}\nu_n(\lambda) \longrightarrow d(0) = d_0. \tag{5.3}$$

For completeness, split the difference from $d(0)$ into $|q| < \varepsilon$ and its complement. The first contribution is at most $\sup_{|q|<\varepsilon}|d(q) - d(0)|$, and the second is at most $2\,\|d\|_\infty\,\nu_n\{|q| \ge \varepsilon\}$. First let $n \to \infty$ and then $\varepsilon \downarrow 0$.

Applying Proposition 5 to x_n gives

$$\int_S d(q(\lambda))\,\mathrm{d}\nu_n(\lambda) \le \kappa_N \mathcal{E}(x_n) \longrightarrow 0,$$

contradicting (5.2) and (5.3). Hence a positive lower bound c exists on $D(A)$.

Admissibility makes $x \mapsto CT(\cdot)x$ a bounded map from the dense subspace $D(A) \subset X$ into $L^2((0,\infty); Y)$. It has a unique bounded extension to X, and the lower bound passes to that extension by density. This proves the stated conclusion. □

6 Scope, use of assumptions, and review points

What the argument claims. The conclusion is the exact implication (H) $\Rightarrow$ (O) in the horizontal-strip question, with arbitrary Hilbert output space and arbitrary normal spectral type. The proof is qualitative: it establishes the existence of a positive observability constant and does not give a useful explicit or optimal value.

Where the assumptions enter.

- Normality provides positive scalar spectral measures and the exact functional-calculus identities used in both parts of the proof.
- Exponential stability supplies $a > 0$, keeping the exposed spectral point strictly inside the left half-plane and making finite-time admissibility sufficient for an infinite-time upper bound.
- The horizontal strip ensures graph-norm smoothing at every fixed complex time with positive real part. It also gives compactness of bounded real-coordinate portions of the spectrum and forces $b/h \to 0$ as $h \to \infty$.
- The full left-half-plane Hautus condition is used at $\zeta = \lambda_0$ in the escape lemma. The filter uses only $\zeta = -1/t$, but that does not replace the full hypothesis in the theorem.
- Graph-norm boundedness of C gives locally bounded holomorphic complex-time observations. Admissibility bounds the real-time observation map and permits its extension to all of X.

Checks that matter. The scalarization in Lemma 2 allows the norming functional to depend on n; no finite-dimensional compactness of Y is used. The smoothing in (3.3) makes the measure family tight and gives a finite second moment for the limit, which is exactly what (3.8) needs. The weak-limit passage uses bounded continuous functions at fixed complex time. The later concentration limit is separate. Finally, (4.6) is applied to the whole vector, so there is no addition of observation lower bounds from separate spectral blocks, where cancellation could otherwise invalidate the proof.

Boundary of the claim. A spectrum confined to a fixed sector with positive opening need not satisfy $b/h \to 0$ at high frequency. Thus the last step here does not establish the general sector version of the question. No such extension is asserted in this draft.

Verification status. This text presents a complete proposed chain of lemmas, rather than an externally established literature result. Independent specialist review should focus in particular on the escape lemma and the passage from real-time observation smallness to fixed complex-time norm smallness. Internal checking has not identified a gap; that is not a substitute for independent mathematical review.

References

[1] J. Daun and B. Jacob, *On the Hautus test for exact observability of normal semigroups*, preprint, arXiv:2608.09407v2 (2026). https://arxiv.org/abs/2608.09407v2. Relevant locations: Section 4 (self-adjoint filtering proof) and Section 6 (horizontal-strip and sector questions).